

Computing Platform for Autonomous Driving (CP4AD)

L2. RTOS #2

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FAQ

- preference_avr8__path_for_Arduino_sdk 오류

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  SERIAL MONITOR
powershell + - [ ] [X] ... ^ X

PS C:\Users\AVEES\Desktop\test\cp4ad-rtos\01. Hello World> make
cd 'out'; sed -i 's/cygpah -ms "C:\Arduino"/cygpah -ms "C:\\Arduino"/g' makefile; make BINDIR="/cygdrive/C/Arduino/hardware/tools/avr/bin"
make[1]: Entering directory '/cygdrive/c/Users/AVEES/Desktop/test/cp4ad-rtos/01. Hello World/out'
cygpah: can't convert empty path
make[2]: Entering directory '/cygdrive/c/Users/AVEES/Desktop/test/cp4ad-rtos/01. Hello World/erika'
ERIKA_FILES=/cygdrive/c/eclipse/plugins/COMEUE~4.201/ee_files
make[3]: Entering directory '/cygdrive/c/Users/AVEES/Desktop/test/cp4ad-rtos/01. Hello World/erika'
*****
* Pulling erika files from:
*****
* ERIKA_FILES=/cygdrive/c/eclipse/plugins/COMEUE~4.201/ee_files
/cygdrive/c/Users/AVEES/Desktop/test/CP4AD~1/0155F9~1.HEL/erika/pull/ee_oscfg_mk.txt:71: *** Provided path 'preference_avr8__path_for_arduino_sdk not defined'
not found.. Stop.
make[3]: Leaving directory '/cygdrive/c/Users/AVEES/Desktop/test/cp4ad-rtos/01. Hello World/erika'
make[2]: *** [makefile:88: mk] Error 2
make[2]: Leaving directory '/cygdrive/c/Users/AVEES/Desktop/test/cp4ad-rtos/01. Hello World/erika'
make[1]: *** [makefile:97: /cygdrive/c/Users/AVEES/Desktop/test/CP4AD~1/0155F9~1.HEL/erika/mk/ee.mk] Error 2
make[1]: Leaving directory '/cygdrive/c/Users/AVEES/Desktop/test/cp4ad-rtos/01. Hello World/out'
make: *** [makefile:12: build] Error 2
```

- make config 이전에 make를 실행 시 발생
- makefile에서 12번째 줄 수정
- "C:\Arduino"/cygpah -> ""/cygpah

FAQ

- 아두이노 업로드 오류
 - 지정된 파일을 찾을 수 없습니다.
-> COM 포트 번호 확인
 - 액세스가 거부되었습니다.
-> 시리얼 모니터 종료 후 다시 업로드 실행
- 그 외
 - 파일 수정 후 “저장” 확인, 저장이 안될 시 오류 발생
 - makefile 각 프로젝트마다 수정 필요

06. Alarm

- OIL에 COUNTER와 ALARM 추가

conf.oil

```
COUNTER counter1 {  
    MINCYCLE = 1;  
    MAXALLOWEDVALUE = 127;  
    TICKSPERBASE = 1;  
};
```

```
ALARM alarm1 {  
    COUNTER = counter1;  
    ACTION = ACTIVATETASK {  
        TASK = Task1;  
    };  
    AUTOSTART = TRUE {  
        ALARMTIME = 5;  
        CYCLETIME = 10;  
    };  
};
```

conf.oil

```
ALARM alarm2 {  
    COUNTER = counter1;  
    ACTION = ACTIVATETASK {  
        TASK = Task2;  
    };  
    AUTOSTART = TRUE {  
        ALARMTIME = 5;  
        CYCLETIME = 20;  
    };  
};
```

06. Alarm

- TimerISR에서 counter1 증가

asw.c

```
ISR2(TimerISR)
{
    static long c = -4;
    IncrementCounter(counter1);
    printfSerial("\n%4ld: ", c++);
}
```

asw.c

```
TASK(Task1)
{
    printfSerial("Task1 Begins...");
    mdelay(3000);
    printfSerial("Task1 Finishes...");
    TerminateTask();
}

TASK(Task2)
{
    printfSerial("Task2 Begins...");
    mdelay(3000);
    printfSerial("Task2 Finishes...");
    TerminateTask();
}
```

06. Alarm

- Alarm을 이용한 주기적 Task 실행

```
Monitor Mode Serial View Mode Text Port COM9 - Arduino Uno(COM9) Baud rate 9600
---- Opened the serial port COM9 ----
.....
...OS Starts...
.....

-4:
-3:
-2:
-1:
0: Task2 Begins...
1:
2:
3: Task2 Finishes...Task1 Begins...
4:
5:
6: Task1 Finishes...
7:
8:
9:
10: Task1 Begins...
11:
12:
13: Task1 Finishes...
14:
15:
16:
17:
18:
19:
20: Task2 Begins...
21:
22:
23: Task2 Finishes...Task1 Begins...
24:
25:
26: Task1 Finishes...
27:
28:
---- Closed the serial port COM9 ----
```

07. Event

conf.oil

```
...
CPU_DATA = AVR8 {
    MULTI_STACK = TRUE;
};
...
EVENT Event1 { MASK = AUTO; };
EVENT Event2 { MASK = AUTO; };
...
TASK Task2 {
    PRIORITY = 2;
    STACK = PRIVATE {
        SIZE = 512;
    };
    SCHEDULE = FULL;
    EVENT = Event1;
    EVENT = Event2;
...

```

asw.c

```
ISR2(ButtonISR)
{
    int a0;
    DisableAllInterrupts();
    if ((PINC & 0x01) != 0) return;
    printfSerial("<BUTTON ISR>");
    a0 = analogRead(A0); // read ADC value
    if (a0 < 50) { // UP
        SetEvent(Task2, Event1);
    } else if (a0 < 200) { // DOWN
        SetEvent(Task2, Event2);
    } else if (a0 < 380) { // LEFT
        ;
    } else if (a0 < 520) { // RIGHT
        ;
    }
    EnableAllInterrupts();
}
```

07. Event

asw.c

```
TASK(Task2)
{
    EventMaskType mask;
    printfSerial("Task2 Begins...");
    printfSerial("Task2 Waits...");
    WaitEvent(Event1 | Event2);
    printfSerial("Task2 Wakes Up...");
    GetEvent(Task2, &mask);
    if (mask & Event1) {
        printfSerial("[Event1]");
        ClearEvent(Event1);
    }
    if (mask & Event2) {
        printfSerial("[Event2]");
        ClearEvent(Event2);
    }
}
```

asw.c

```
    printfSerial("Task2 Finishes...");
    TerminateTask();
}

ISR2(TimerISR)
{
    static long c = -4;
    if (c == 0) {
        ActivateTask(Task1);
    }
    printfSerial("\n%4ld: ", c++);
}
```

07. Event

- ClearEvent는 왜 필요한가?
- 우선순위 반대의 경우 스케줄링

```
Monitor Mode Serial View Mode Text Port COM34 - Arduino Uno(COM34)
.....
...OS Starts...
.....

-4:
-3:
-2:
-1:
0: Task1 Begins...
1:
2:
3: Task2 Begins...Task2 Waits...
4:
5:
6: Task1 Finishes...
7: <BUTTON ISR>Task2 Wakes Up...[Event1]Task2 Finishes...
8:
9:
10:
```

```
Monitor Mode Serial View Mode Text Port COM34 - Arduino Uno(COM34)
---- Opened the serial port COM34 ----

.....
...OS Starts...
.....

-4:
-3:
-2:
-1:
0: Task1 Begins...
1:
2:
3: Task2 Begins...Task2 Waits...
4:
5:
6: Task1 Finishes...
7: <BUTTON ISR>Task2 Wakes Up...[Event2]Task2 Finishes...
8:
9:
```

08. Alarm SetEvent

- Alarm의 SetEvent Action

conf.oil

```
COUNTER counter1 {
    MINCYCLE = 1;
    MAXALLOWEDVALUE = 127;
    TICKSPERBASE = 1;
};
ALARM alarm1 {
    COUNTER = counter1;
    ACTION = SETEVENT {
        TASK = Task2;
        EVENT = Event1;
    };
    AUTOSTART = TRUE {
        ALARMTIME = 15;
        CYCLETIME = 20;
    };
};
```

asw.c

```
ISR2(TimerISR)
{
    static long c = -4;
    if (c == 0) {
        ActivateTask(Task1);
    }
    IncrementCounter(counter1);
    printfSerial("\n%4ld: ", c++);
}
```

08. Alarm SetEvent

- Alarm의 SetEvent Action

```
Monitor Mode Serial View Mode Text Port COM9 - Arduino Uno(COM9) Baud rate 9600
---- Opened the serial port COM9 ----

.....
...OS Starts...
.....

-4:
-3:
-2:
-1:
0: Task1 Begins...
1:
2:
3: Task2 Begins...Task2 Waits...
4:
5:
6: Task1 Finishes...
7:
8:
9:
10: Task2 Wakes Up...[Event1]Task2 Finishes...
11:
12:
13:
---- Closed the serial port COM9 ----
```

09-1. Loss

- 공유 자원 접근으로 인한 Data Loss 실험
- Task1: Low priority, AUTOSTART
- Task2: High priority, 1 ms 주기 실행
- 공유 전역 변수 "shared"
- Task1에서 100만번 shared++
- Task1이 한 번 실행되는 동시에 Task2 1000번 실행하며 shared++
- 양 쪽 태스크에서 더한 숫자가 모두 유지가 되는지 확인

09-1. Loss

- 공유 전역변수 추가 및 타이머 주기 수정

asw.c

```
volatile unsigned long shared = 0;
...

TASK(Task1)
...
    for (i = 0; i < 1000000; i++) {
        shared++;
    }
...

TASK(Task2)
...
    if (i < 1000) {
        shared++;
    }
...
```

bws.cpp

```
#define TIMER1_US 1000U /* 1 msec*/
```

09-1. Loss

- Resource를 이용해서 Integrity Loss 문제 해결 필요

```
Monitor Mode Serial View Mode Text Port COM9 - Arduino Uno(COM9)
---- Opened the serial port COM9 ----

.....
...OS Starts...
.....
Task1 Begins...
Added 1000 to shared
Added 1000000 to shared
counter = 1000792
Task1 Finishes...
---- Closed the serial port COM9 ----
```

$$1000000 + 1000 = 1000792 (???)$$

09-2. No Loss

- OSEK의 RESOURCE 기능을 이용하여 Data Loss 문제 해결

asw.c

```
TASK(Task1)
...
    GetResource(S1);
    shared++;
    ReleaseResource(S1);
...

TASK(Task2)
...
    GetResource(S1);
    shared++;
    ReleaseResource(S1);
...
```

conf.oil

```
TASK Task1 {
    PRIORITY = 1;
    STACK = SHARED;
    SCHEDULE = FULL;
    AUTOSTART = TRUE;
    ACTIVATION = 1;
    RESOURCE = S1;
};

RESOURCE S1 {
    RESOURCEPROPERTY = STANDARD;
};

...
```

Task2 에도
동일하게 추가

09-2. No Loss

- Data Integrity 문제 해결

```
Monitor Mode Serial View Mode Text Port COM9 - Arduino Uno(COM9)
---- Opened the serial port COM9 ----

.....
...OS Starts...
.....
Task1 Begins...
Added 1000 to shared
Added 1000000 to shared
counter = 1001000
Task1 Finishes...
---- Closed the serial port COM9 ----
|
```

$$1000000 + 1000 = 1001000 (!!!)$$

10. Priority Inversion

• mutex.c

Waiting/Wakeup 을 위해
Event 지정 필요

```
#include "ee.h"
#include "bsw.h"
#include "mutex.h"
```

```
void InitMutex(MutexType *mutex, EventMaskType event)
{
    mutex->flag = UNLOCKED;
    mutex->waiting_task = 0;
    mutex->event = event;
}
```

```
void GetMutex(MutexType *mutex)
{
    if (mutex->flag == LOCKED) {
        printfSerial("  --> BLock");
        GetTaskID(&(mutex->waiting_task));
        WaitEvent(mutex->event);
    }
    mutex->flag = LOCKED;
}
```

```
void ReleaseMutex(MutexType *mutex)
{
    if (mutex->flag == LOCKED) {
        mutex->flag = UNLOCKED;
        if (mutex->waiting_task != 0) {
            SetEvent(mutex->waiting_task, mutex->event);
        }
    }
}
```

**PCP 없이 Mutex 사용할 경우 문제점을
확인하기 위한 Dummy 구현**

10. Priority Inversion

- mutex.h

```
#ifndef MUTEX_H_
#define MUTEX_H_

#define LOCKED    1
#define UNLOCKED  0

typedef struct _MutexType {
    int flag;
    EventMaskType event;
    TaskType waiting_task;
} MutexType;

void InitMutex(MutexType *mutex, EventMaskType event);

void GetMutex(MutexType *mutex);

void ReleaseMutex(MutexType *mutex);

#endif /* MUTEX_H_ */
```

10. Priority Inversion

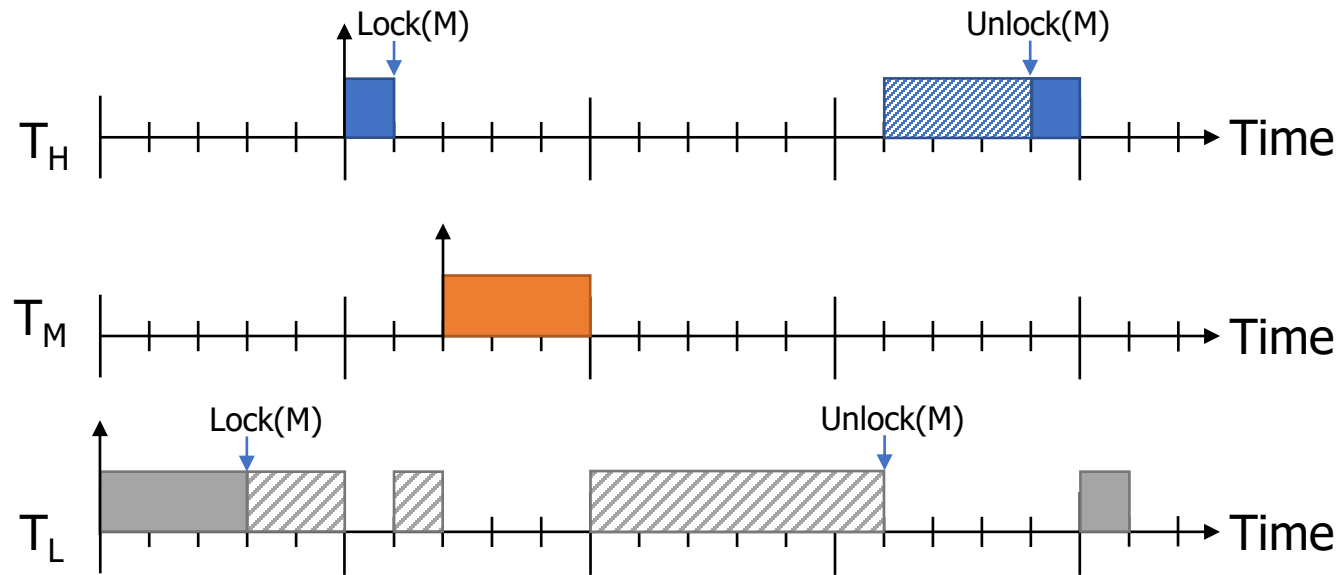
- Mutex 선언
- Timer ISR 이용
 - Mutex 초기화
 - Task Activation
- Mutex의 동작 확인

```
MutexType M1;
MutexType M2;

ISR2(TimerISR)
{
    static long c = -5;
    printfSerial("\n%4ld: ", ++c);
    if(c == -4) {
        InitMutex(&M1, Event1);
    }
    else if (c == 0) {
        ActivateTask(TaskL);
    }
    else if (c == 5) {
        ActivateTask(TaskH);
    }
    else if (c == 7) {
        ActivateTask(TaskM);
    }
}
```

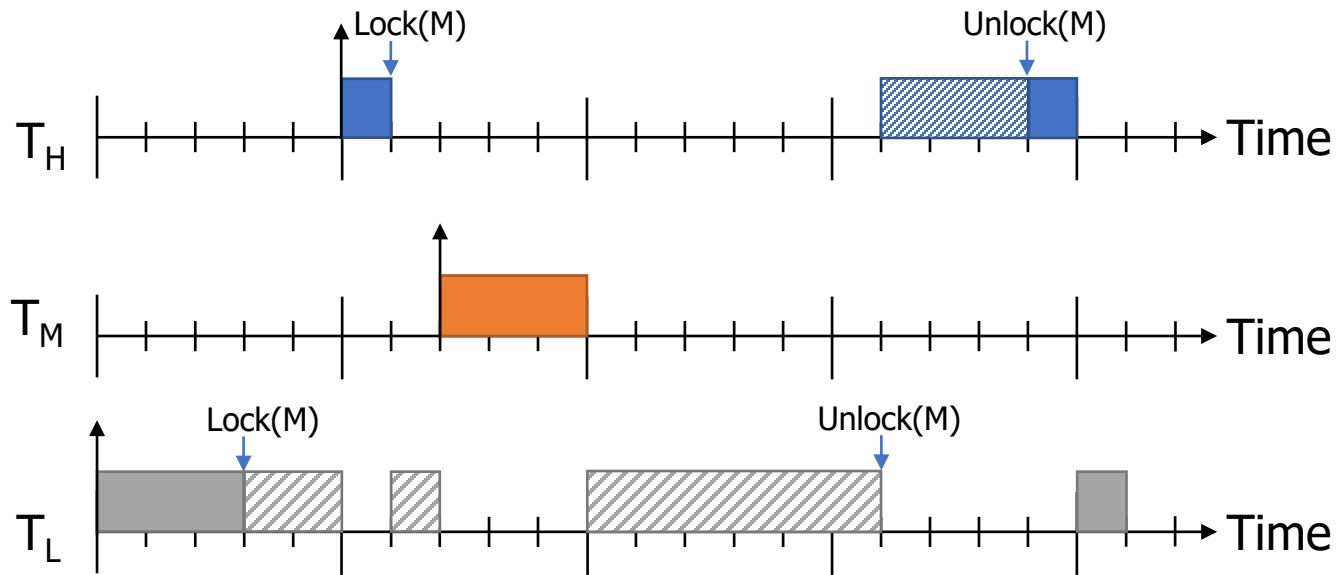
10. Priority Inversion

- Mutex를 활용하여 아래 스케줄 재현하기



10. Priority Inversion

- Mutex 사용, Task 선언 (Priority, Event)

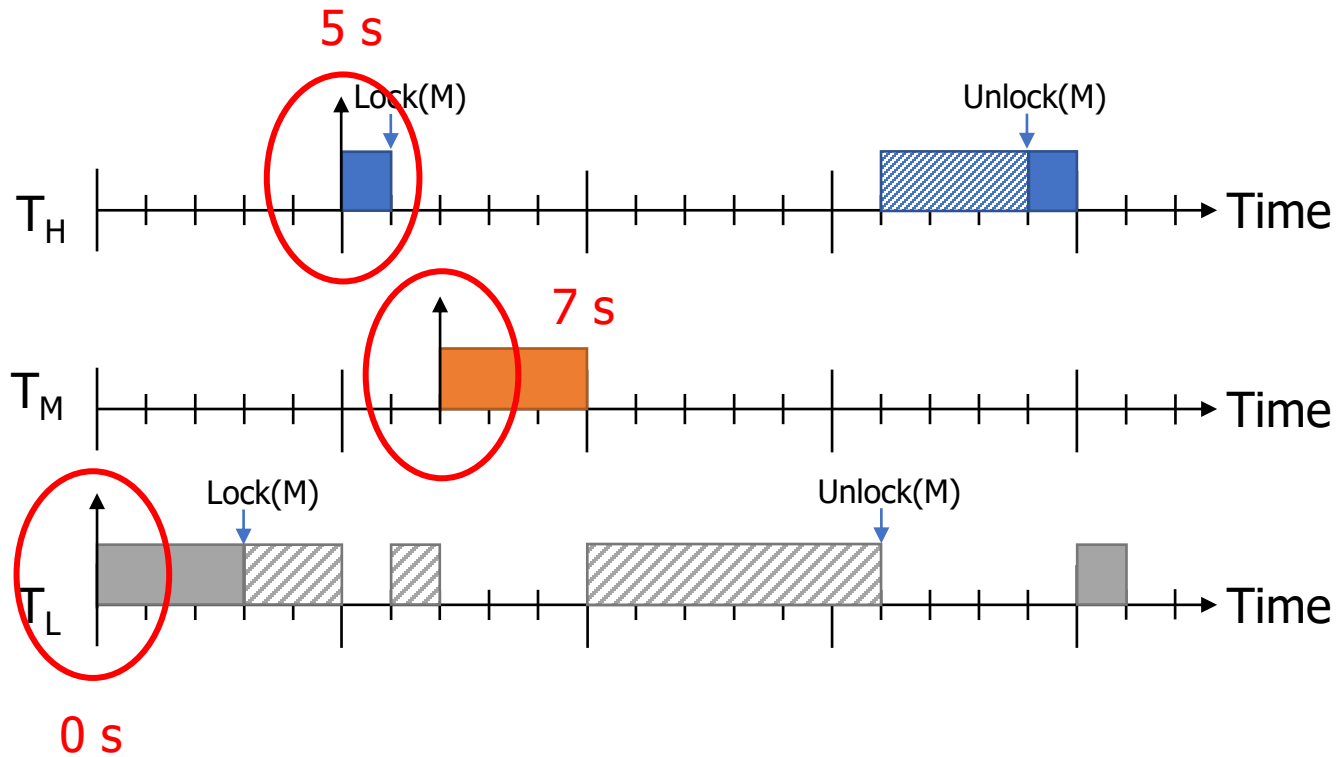


```
APPDATA myApp {  
    ...  
    APP_SRC = "mutex.c";  
};  
  
EVENT Event1 { MASK = AUTO; };  
  
TASK TaskH {  
    PRIORITY = 3;  
    ...  
    EVENT = Event1;  
};  
  
TASK TaskM {  
    PRIORITY = 2;  
    ...  
};  
  
TASK TaskL {  
    PRIORITY = 1;  
    ...  
    EVENT = Event1;  
};
```

클 수록 높은
우선순위

10. Priority Inversion

- Task 시작 지점

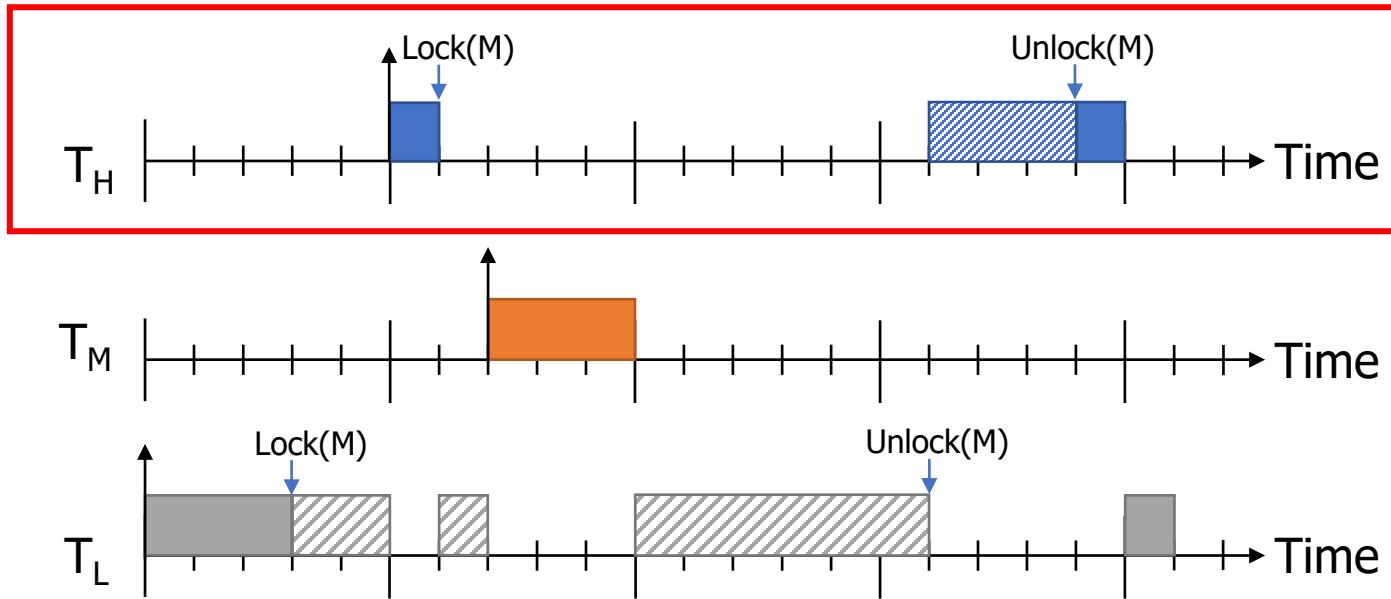


MutexType M1;

```
ISR2(TimerISR)
{
    static long c = -5;
    printfSerial("\n%4ld: ", ++c);
    if(c == -4) {
        InitMutex(&M1, Event1);
    }
    else if (c == 0) {
        ActivateTask(TaskL);
    }
    else if (c == 5) {
        ActivateTask(TaskH);
    }
    else if (c == 7) {
        ActivateTask(TaskM);
    }
}
```

10. Priority Inversion

- Task H (Priority 3)



```
TASK(TaskH)
{
    printfSerial("<TaskH begins.> ");
    mdelay(1000);

    printfSerial("TaskH : Try Lock(M1). ");
    GetMutex(&M1);
    printfSerial("TaskH : Get Lock(M1). ");

    mdelay(3000);

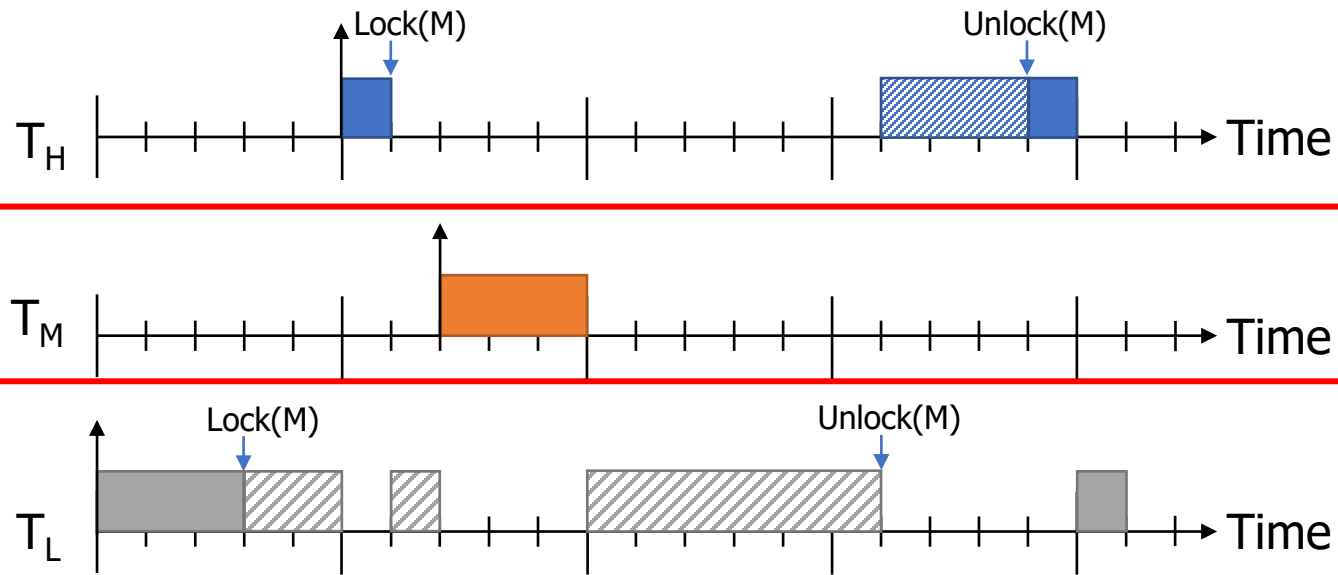
    printfSerial("TaskH : Release Lock(M1).");
    ReleaseMutex(&M1);

    mdelay(1000);

    printfSerial("<TaskH ends.> ");
    TerminateTask();
}
```

10. Priority Inversion

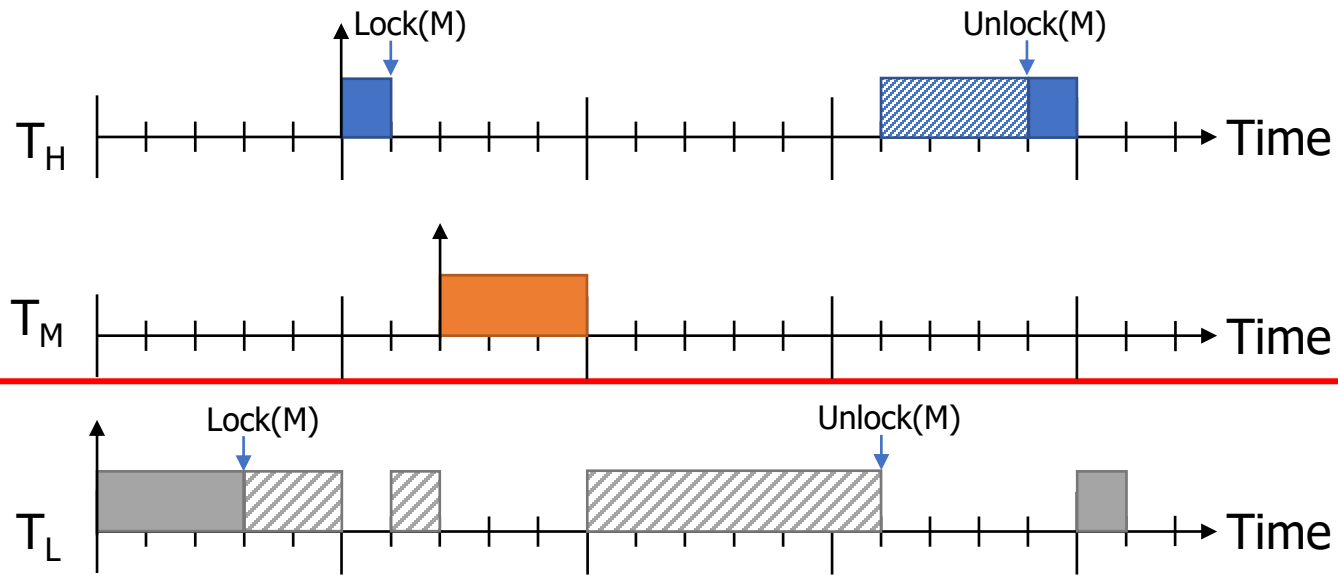
- Task M (Priority 2)



```
TASK(TaskM)
{
    printfSerial("<TaskM begins.> ");
    mdelay(3000);
    printfSerial("<TaskM ends.> ");
    TerminateTask();
}
```

10. Priority Inversion

- Task L (Priority 1)



```
TASK(TaskL)
{
    printfSerial("<TaskL begins.> ");
    mdelay(3000);

    printfSerial("TaskL : Try Lock(M1). ");
    GetMutex(&M1);
    printfSerial("TaskL : Get Lock(M1). ");

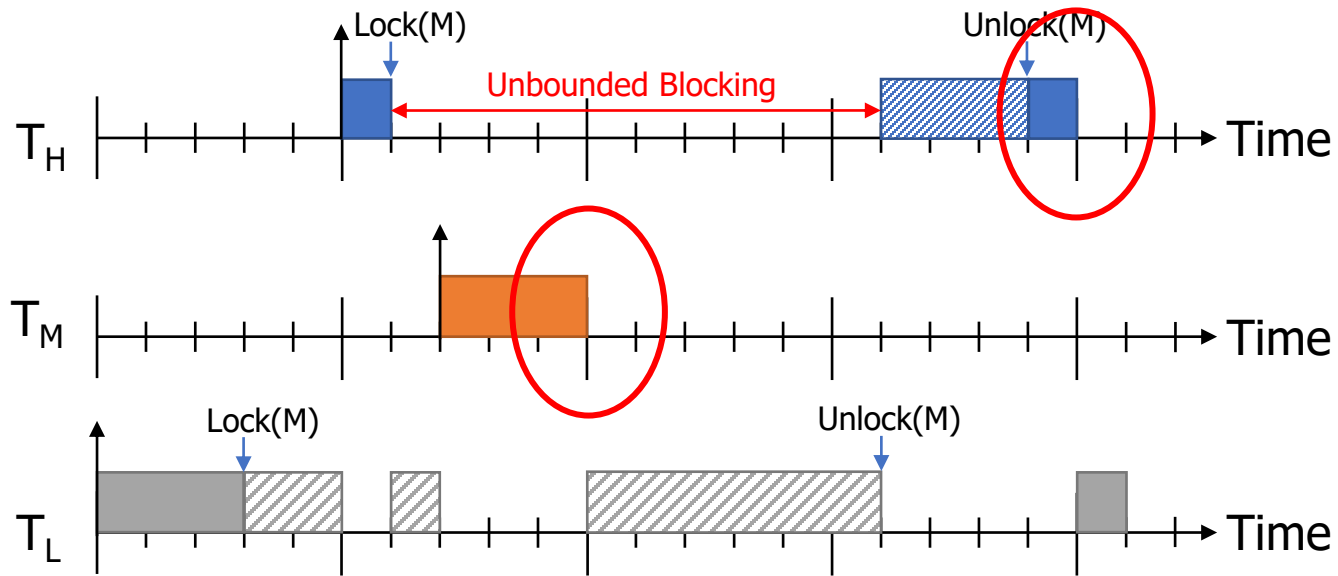
    mdelay(9000);

    printfSerial("TaskL : Release Lock(M1).");
    ReleaseMutex(&M1);

    mdelay(1000);
    printfSerial("<TaskL ends.> ");
    TerminateTask();
}
```

10. Priority Inversion

- Priority Inversion 발생



```
Monitor Mode Serial View Mode Text Port COM9 - Arduino Uno(COM9)

---- Opened the serial port COM9 ----

.....
...OS Starts...
.....

-4:
-3:
-2:
-1:
0: <TaskL begins.>
1:
2:
3: TaskL : Try Lock(M1). TaskL : Get Lock(M1).
4:
5: <TaskH begins.>
6: TaskH : Try Lock(M1). --> BLock
7: <TaskM begins.>
8:
9:
10: <TaskM ends.>
11:
12:
13:
14:
15:
16: TaskL : Release Lock(M1). TaskH : Get Lock(M1).
17:
18:
19: TaskH : Release Lock(M1).
20: <TaskH ends.>
21: <TaskL ends.>
22:
23:

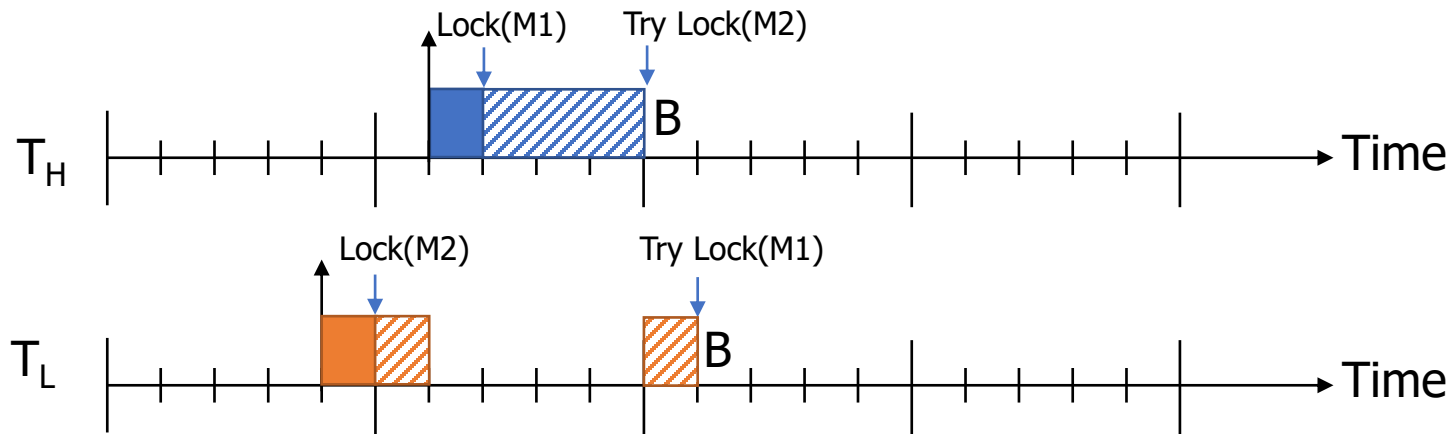
---- Closed the serial port COM9 ----
```

Unbounded Blocking

11. Deadlock

- Mutex를 활용하여 Deadlock 재현 직접 진행해보기

1. asw.c 파일만 수정
2. 그림 기반으로 10번 실습 되짚어가며 진행



```
Monitor Mode Serial
---- Opened the serial port COM9 ----

.....
...OS Starts...
.....

-4:
-3:
-2:
-1:
0:
1:
2:
3:
4: <TaskL begins.>
5: TaskL : Try Lock(M2). TaskL : Get Lock(M2).
6: <TaskH begins.>
7: TaskH : Try Lock(M1). TaskH : Get Lock(M1).
8:
9:
10: TaskH : Try Lock(M2). --> BLock
11: TaskL : Try Lock(M1). --> BLock
12:
13:      Deadlock
14:
15:
---- Closed the serial port COM9 ----
```

Questions

